ERA 5 calibration to Elexon power

S. Gomez[1]

[1]School of Mathematics, University of Edinburgh

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Overview

Updates

New

- Probabilistic Scenarios article accepted in Spatial Statistics
- Project goal and summaries
- Spatiotemporal model for calibration

Project Goal

Project Goal

- Correct ERA5 wind power biases using observed generation data.
- Reduce error metrics (RMSE/MAE/bias).
- Ensure spatially consistent estimates that generalise out-of-sample.
- Capture low-wind events: frequency, duration, and spatial extent.

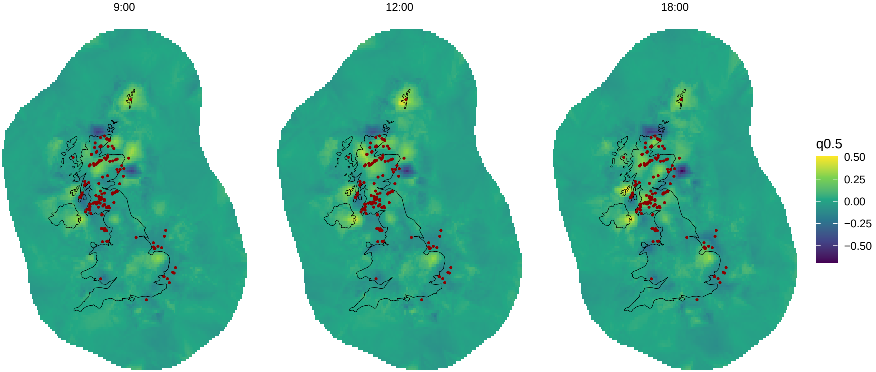
Spatiotemporal calibration model

Model overview

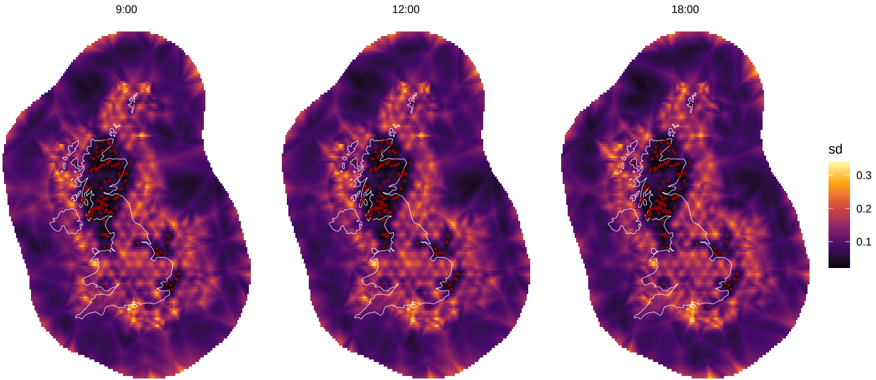
$$\mathbb{E}(p_{it}^{\text{calib}}) = \alpha + \beta_k + \beta_m + \beta_h + s_p(\hat{p}_{it}) + s_w(w_{it}) + u_{it}, \tag{1}$$

- Calibrated power: corrected output at site *i*, time *t*.
- α: intercept.
- Fixed effects $\beta_k, \beta_m, \beta_h$: location/season/time structure.
- ERA5 correction s_p: bias adjustment of ERA5 + generic PC estimate.
- Wind effect s_w: nonlinear wind–power correction.
- Spatiotemporal effect u_{it}: Spatial component + AR1 error.

Estimated spatial field

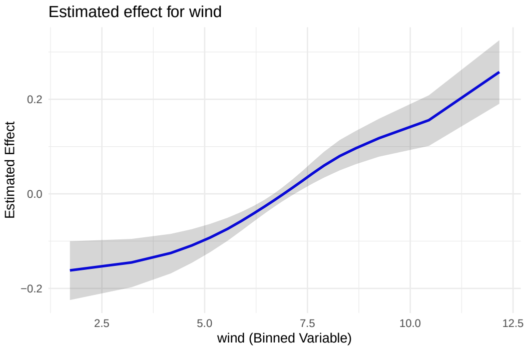


Standard deviation of the spatial field

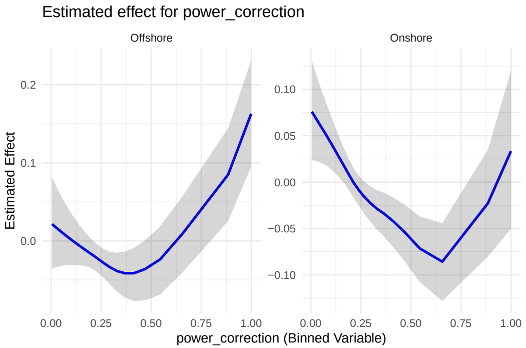


Random Effects

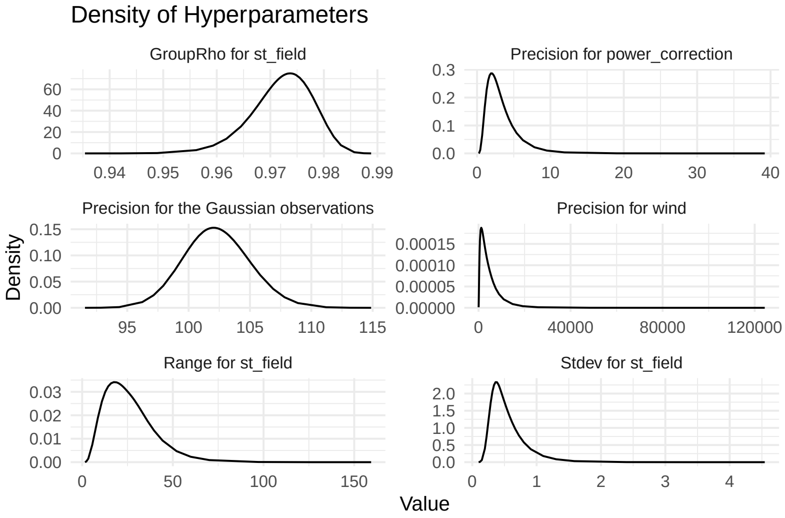
Correction by wind speed



Correction by power regime



Hyperparameters



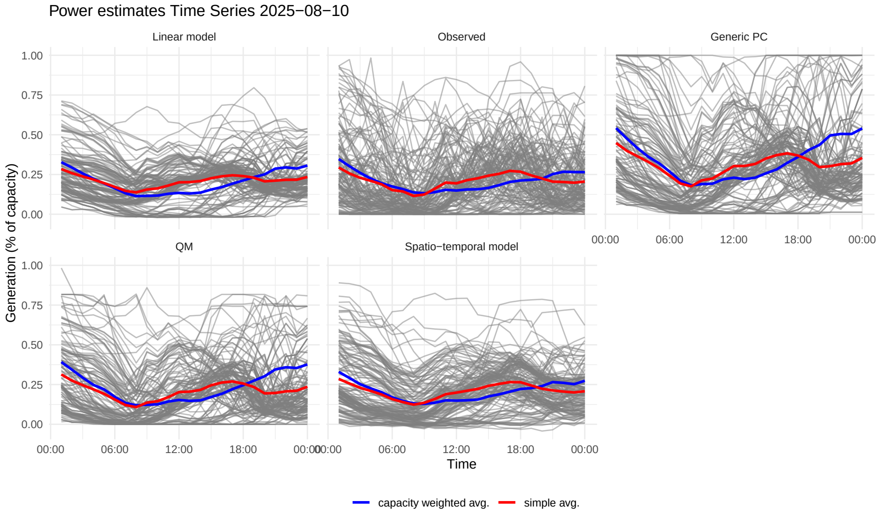
Comparison scores

model	RMSE	MAE	Bias
Generic PC	0.213	0.153	1.02e-01
QM	0.154	0.108	-2.50e-04
Linear model	0.137	0.101	-1.15e-17
Spatio-temporal model	0.094	0.065	8.57e-11

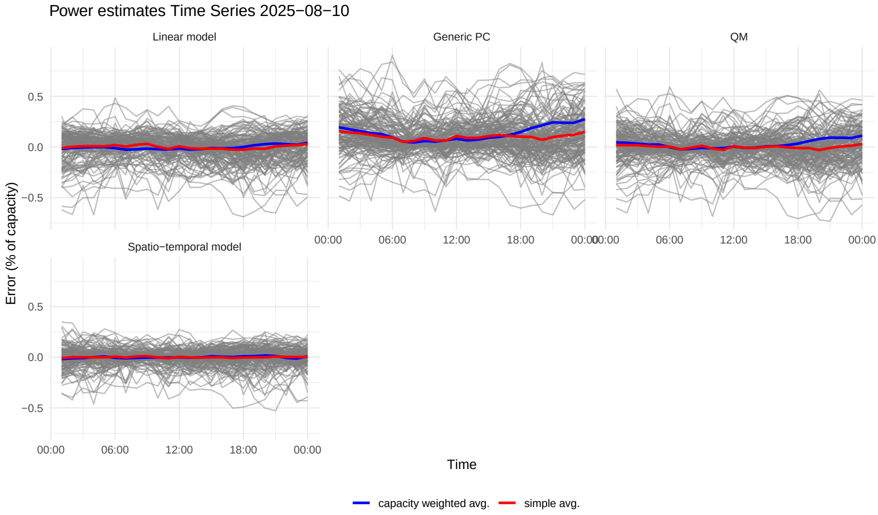
Scores by Offshore / Onshore

tech_typ	model	RMSE	MAE	Bias
Wind Offshore	Generic PC	0.240	0.161	1.44e-01
Wind Offshore	QM	0.142	0.087	4.60e-02
Wind Offshore	Linear model	0.112	0.077	4.33e-16
Wind Offshore	Spatio-temporal model	0.065	0.040	-3.78e-03
Wind Onshore	Generic PC	0.204	0.151	8.77e-02
Wind Onshore	QM	0.159	0.114	-1.58e-02
Wind Onshore	Linear model	0.144	0.109	-1.61e-16
Wind Onshore	Spatio-temporal model	0.101	0.073	1.27e-03

Estimates by hour

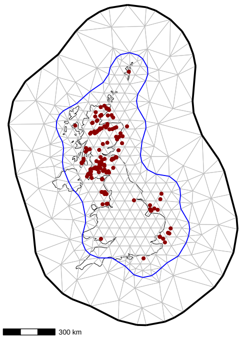


Errors by hour

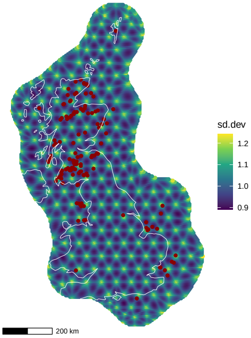


Model Mesh

Mesh used for spatial calibration model

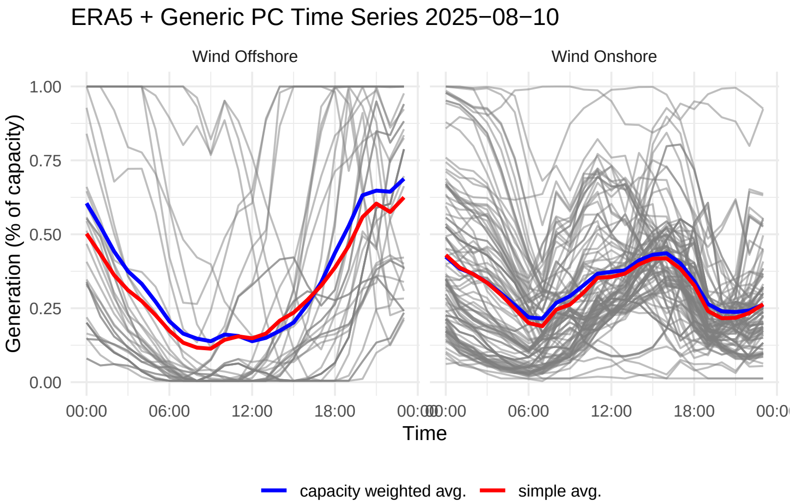


Mesh assessment

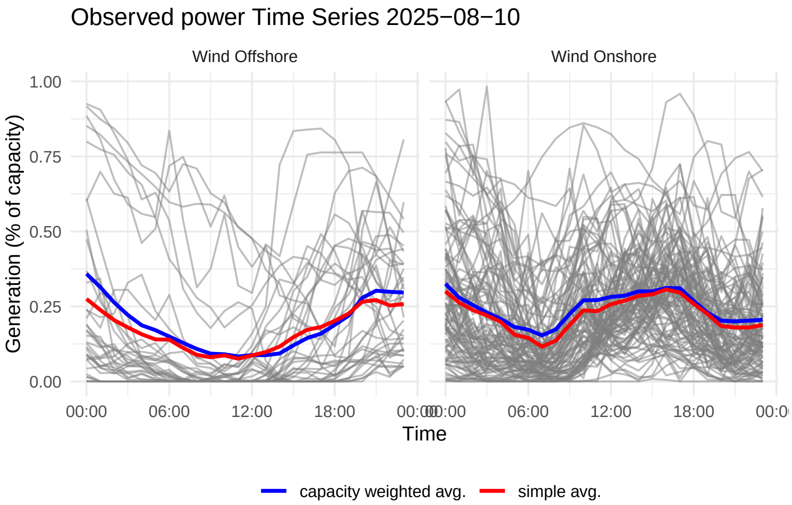


Time series of data and generic PC estimate

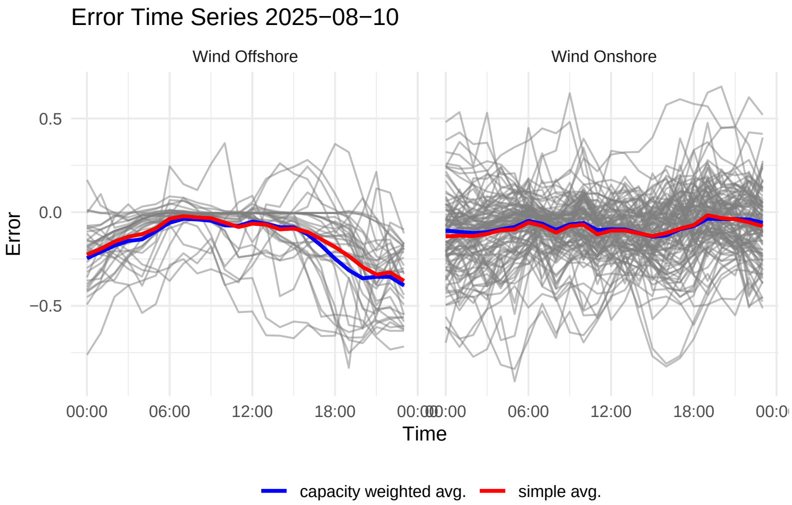
ERA5 data and generic PC estimate



Observed data

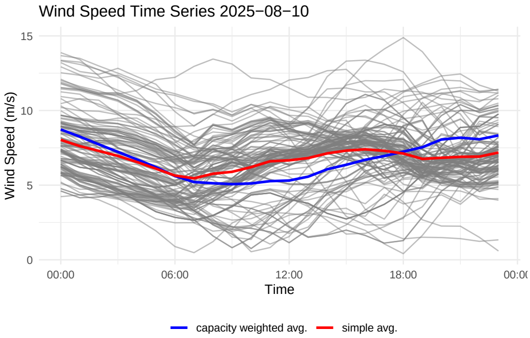


Error of generic PC estimate

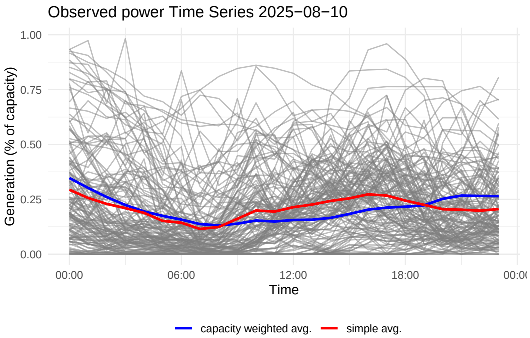


ERA5 wind speed

Wind speed at wind farm locations

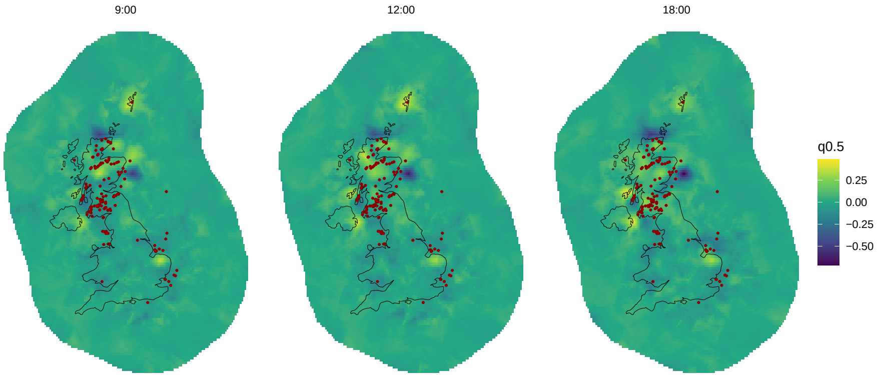


Observed generation

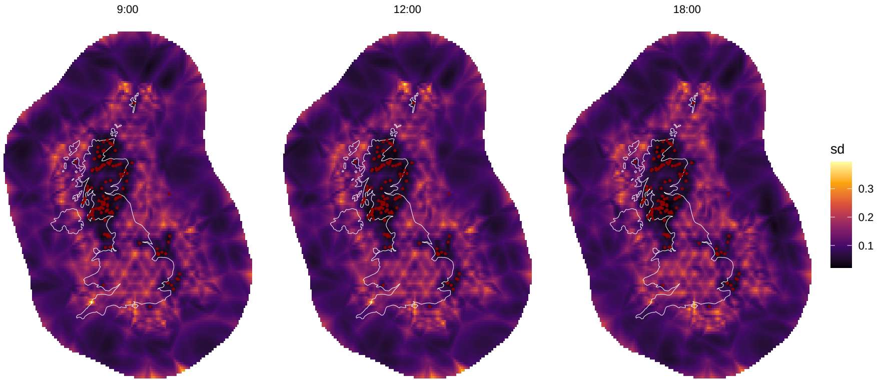


Same day previous year

Estimated spatial field



Standard deviation of the spatial field



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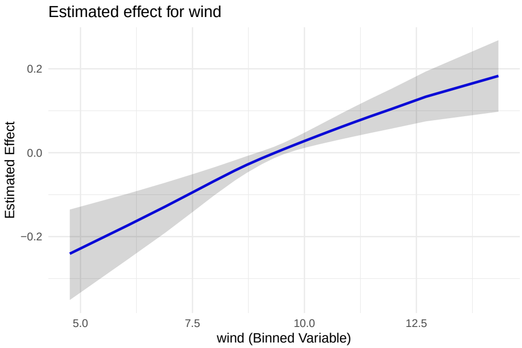
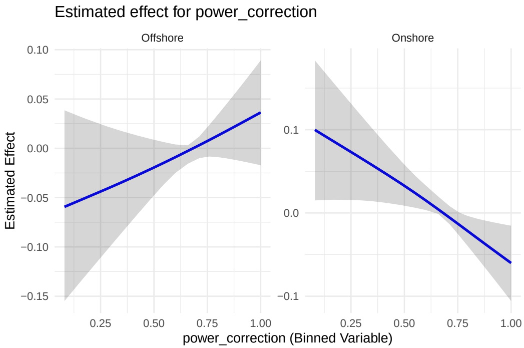
Correction by wind speed



Correction by power regime



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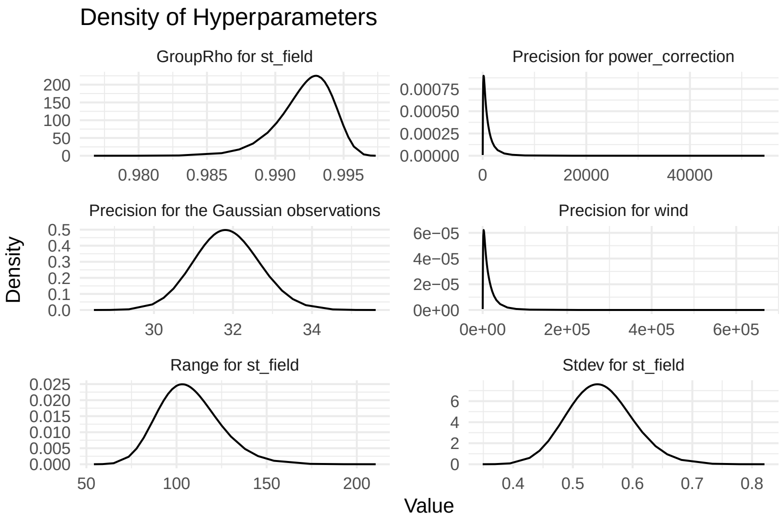


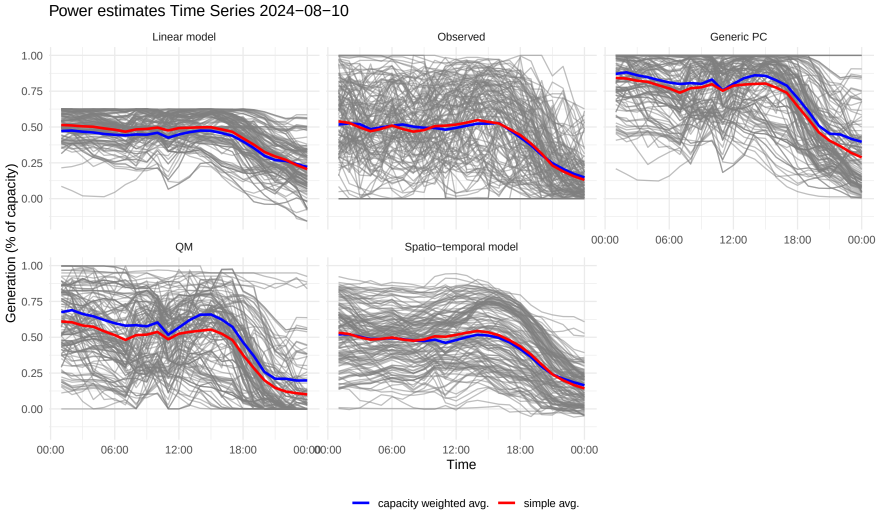
Comparison scores

model	RMSE	MAE	Bias
Generic PC	0.385	0.299	2.50e-01
QM	0.314	0.239	-1.23e-04
Linear model	0.251	0.199	3.38e-17
Spatio-temporal model	0.170	0.125	1.08e-09

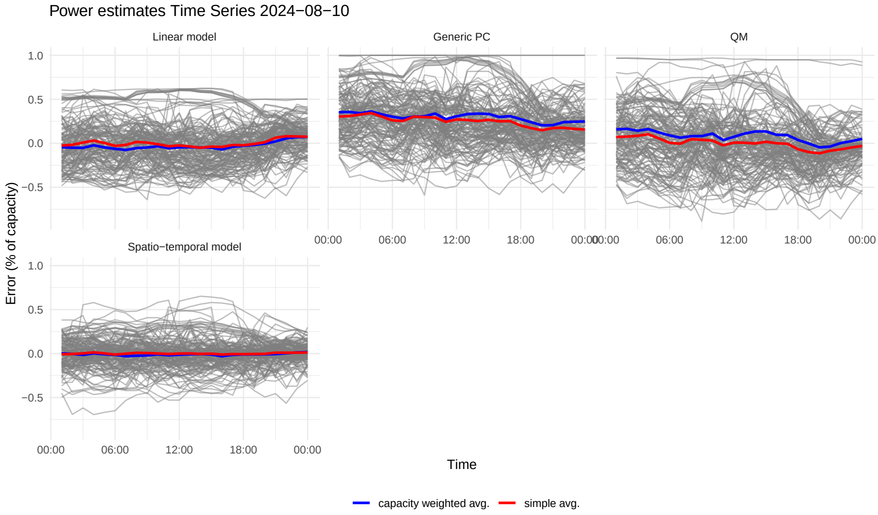
Scores by Offshore / Onshore

tech_typ	model	RMSE	MAE	Bias
Wind Offshore	Generic PC	0.449	0.369	3.61e-01
Wind Offshore	QM	0.341	0.236	1.47e-01
Wind Offshore	Linear model	0.211	0.162	7.09e-17
Wind Offshore	Spatio-temporal model	0.098	0.072	-4.90e-04
Wind Onshore	Generic PC	0.361	0.276	2.13e-01
Wind Onshore	QM	0.305	0.240	-4.88e-02
Wind Onshore	Linear model	0.263	0.211	2.16e-17
Wind Onshore	Spatio-temporal model	0.187	0.143	1.62e-04

Estimates by hour

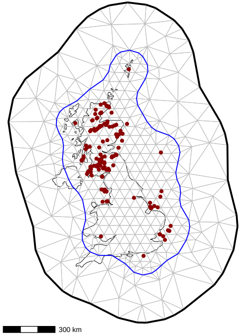


Errors by hour

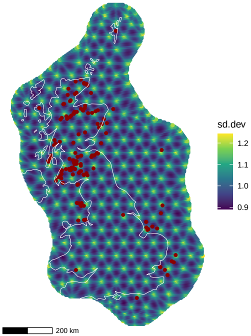


Model Mesh

Mesh used for spatial calibration model

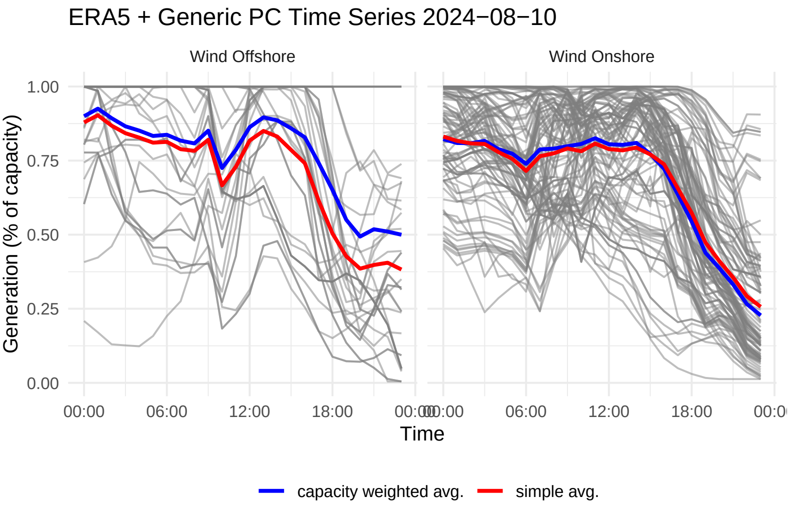


Mesh assessment

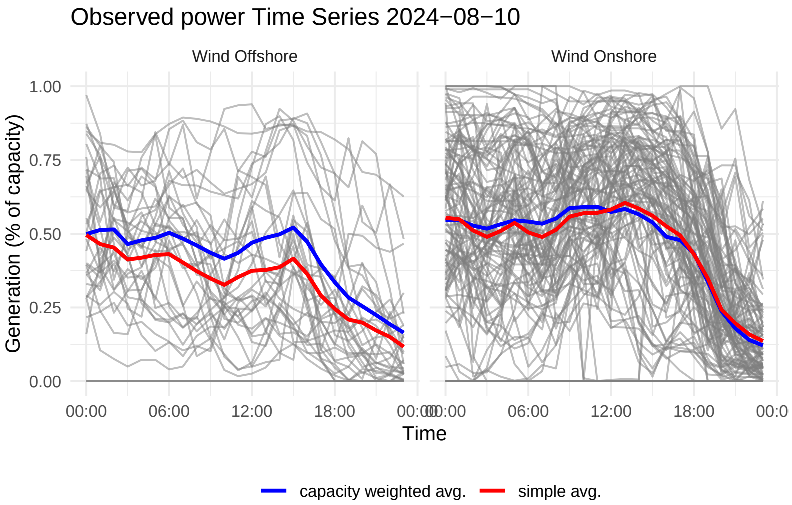


Time series of data and generic PC estimate

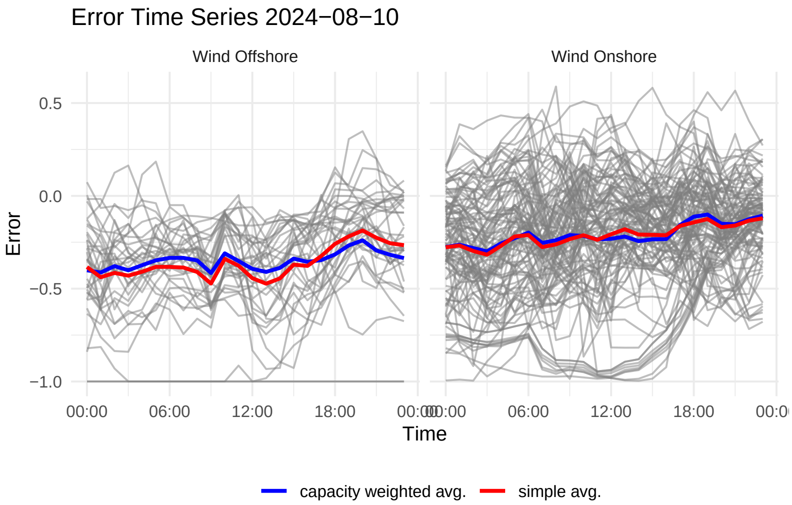
ERA5 data and generic PC estimate



Observed data

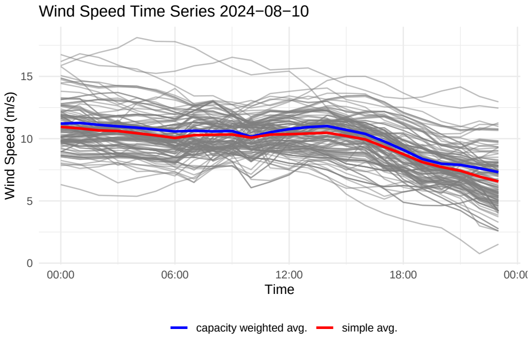


Error of generic PC estimate

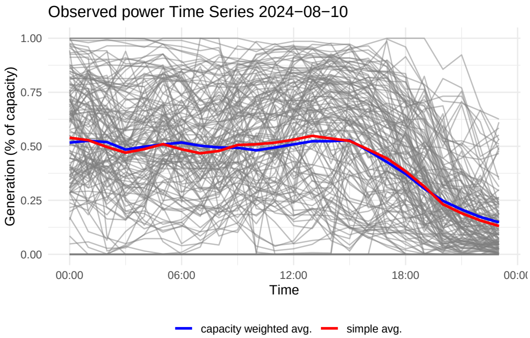


ERA5 wind speed

Wind speed at wind farm locations



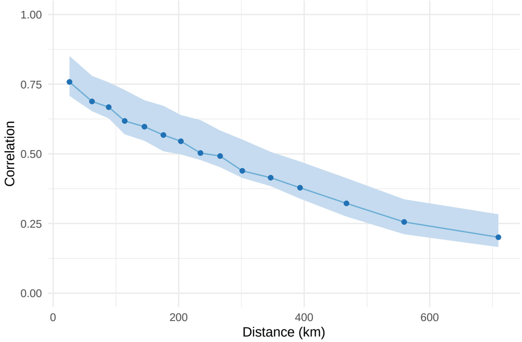
Observed generation



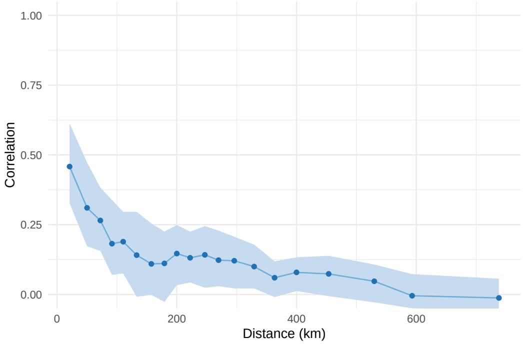
Appendix

Spatial correlation

Correlation for Elexon generation



Correlation for error after calibration



Calibration of generic power curve estimate

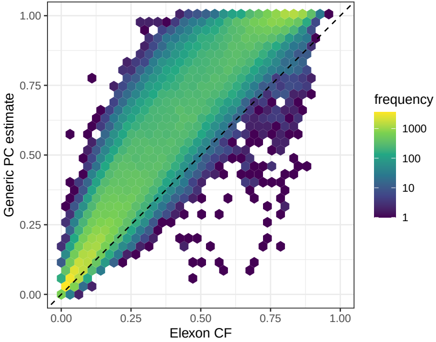
GB level aggregation

- Nearest ERA5 grid for wind speed
- Vertical interpolation using a power law
- Generic Power Curve (onshore, offshore)
- Normalised ower aggregated at GB level

$$p_{\text{tot},t} = \sum_i p_{it} / \sum_i C_i$$

- Hourly resolution
- Potential generation (ex curtailment)

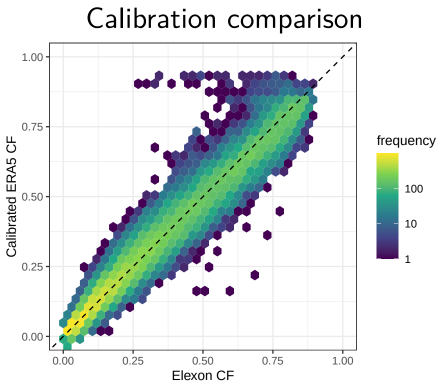
Scatter aggregated power



Linear model

- Using step AIC criteria selection:
  - k: Type (offshore / onshore)
  - m: month, h: hour
  - w_t: wind speed

$$p_t^{calib} = \alpha_{k,m,h} + \beta_{k,m}\hat{p}_t + \sum_{j=1}^{3} \phi_j w_t^3$$

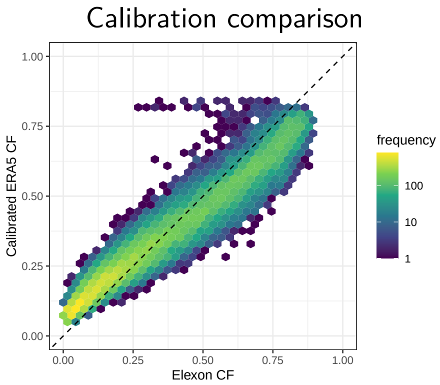


Autoregressive error

- Using step AIC criteria selection:
  - k: Type (offshore / onshore)
  - m: month, h: hour
  - w_t: wind speed
  - u_t: AR1

$$p_t^{calib} = \alpha_0 + \sum \alpha_e + \beta_k \hat{p}_t + f_w(w_t) + u_t$$

where α_0 is the intercept, the α_e $e \in k, m, h$ represent random effects (cyclic for month and hour), β_k is a random slope, f_w is smooth effect (RW2).

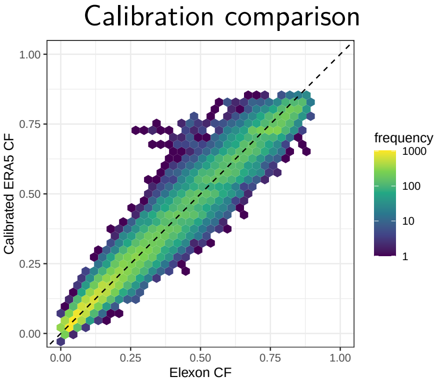


Hierarchical Bayesian model

- Turning muliplicative corrections into smooth functions

$$p_t^{calib} = \alpha + \beta_k + \beta_m + \beta_h + s_p(p_t) + s_w(w_t)$$

where β_k, β_m, β_h are random effects and s_p and s_w are smooth functions

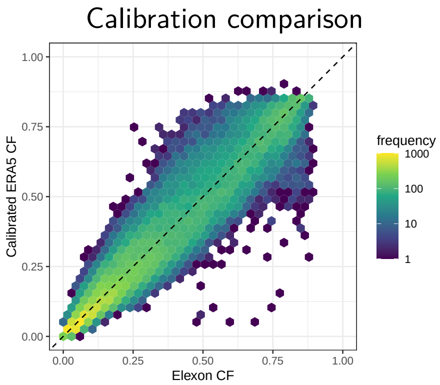


Quantile Mapping

- Quantile mapping aims to replicate observed distribution

$$p_t^{calib} = F_{\text{obs}}(F_{\text{mod}}^{-1}(p_t))$$

where F_{mod} and F_{obs} are the CDF of the power estimate and observed values



Comparison scores

model	RMSE	MAE	Bias
Generic PC	0.200	0.164	1.63e-01
AR inlabru	0.092	0.077	3.16e-04
AR1	0.075	0.062	8.57e-05
QM	0.075	0.054	-2.60e-05
Linear	0.052	0.038	1.39e-19
Linear inlabru	0.049	0.036	-2.61e-10

Power curve modelling

Power curve modelling

Components

- Power curve estimate from data
- Probability of zero generation
- Penalisation to make power curve resemble manufaturer's PC

Using a three-likelihood approach for power curve estimation:

- **1st likelihood:** Observed power is modelled as a smooth function of wind speed
- **2nd likelihood:** Zero inflated component
- **3rd likelihood** Penalisation towards a generic PC

Power model overview

1. $\pi_{it} = \pi(w, i) = \Pr(P_{it} = 0)$ depends of wind speed and location

$$L^{(0)}(\theta^0|\mathbf{p}, \mathbf{w}, \mathbf{i}) = \prod_t \left[\pi_i(w_{it}, i)^{\mathbb{1}(P_{it}=0)} \times (1 - \pi_i(w_{it}, i))^{\mathbb{1}(P_{it}>0)}\right]$$

2. Generation when $P_{it} > 0$ also depends on wind speed / location. Learned as a smooth function from data: $s^{\mathrm{pc}}(w, i) \in (0, 1)$

$$s^{\mathrm{pc}}(w_{it}, i) = \mathsf{E}(P_{it}|P_{it} > 0, w_{it}, i) = \mu^+_{it}$$
$$L^{(\beta)}(\theta^{\mathrm{pc}}|\mathbf{p}, \mathbf{w}, \mathbf{i}) = \prod_{t:P_{it}>0} \mathsf{Beta}\left(P_{it}|\mu^+_{it}, \phi\right),$$

where θ^0 and θ^{pc} represents the parameters of the smooth functions, $\pi(w, i)$ and s^{pc}, and ϕ the precision of the Beta distribution.

Penalisation component

3. Finally, we penalise s^{pc} to be close to our generic power curve estimate, $\hat{PC}_i(w)$: $d_{it} = |\hat{PC}_i(w_{it}) - s^{\text{pc}}(w_{it}, i)|$ should be small.

- Or maximise the negative of $\tau_{\text{ps}} \sum d_{it}^2$ for a certain penalisation parameter τ_{ps}.
- Adding that penalty term is similar to assume $d_{it} \sim N(0, 1/\tau_{\text{ps}}^{-1})$

$$\max_{\theta} \quad l^{(0)}(\theta) + l^{(\beta)}(\theta) - \tau_{\text{ps}} \sum_{i,t} d_{it}^2$$

where $\theta = (\theta^0, \theta^{\text{pc}}, \tau_{\text{ps}})$ represents the parameters of both likelihoods, plus the penalisation parameter.

Power curve model equations

Model is a Zero Inflated Beta (ZIB) with penalisation component:

Zero–probability model (Bernoulli)

$$\eta_{it}^{(0)} = \alpha_{\text{zero}} + f_{\text{zero}}(w_{it},\, i)\,,$$

$$Z_{it} \mid \eta_{it}^{(0)} \sim \text{Bernoulli}(\pi_{it}), \qquad \pi_{it} = \frac{1}{1 + \exp\left\{-\eta_{it}^{(0)}\right\}}\,,$$

Positive–output model (Beta)

$$\eta_{it}^{(\beta)} = \alpha_{\text{pc}} + f_{\text{pc}}(w_{it},\, i)\,,$$

$$P_{it}^{\mathsf{obs}} \mid \eta_{it}^{(\beta)} \sim \text{Beta}(\mu_{it}^{+}, \phi), \qquad \mu_{it}^{+} = \frac{1}{1 + \exp\left\{-\eta_{it}^{(\beta)}\right\}}\,,$$

Pseudo–likelihood model (Gaussian)

$$\hat{PC}_i(w_{it}) \mid \eta_{it}^{(\beta)} \sim \mathcal{N}\left(\eta_{it}^{(\beta)}, \tau_{\text{ps}}^{-1}\right), \qquad \tau_{\text{ps}} = \text{precision or penalisation}\,.$$

Details on smooth components

- The smooth components $f_{\text{zero}}(w, i)$ $f_{\text{pc}}(w, i)$ is modelled as a 1D SPDE over wind speed with a uniform correlation matrix across locations $i = 1, \ldots, I$.

$$f(w, i) = \sum_{i=1}^{I} \beta_i \psi_k(w),$$

where $\psi_k(w)$ are basis functions defined over wind speed and β_i are coefficients The correlation structure of $f(w, i)$ assumes a Matern covariance function: Let w_s, w_h represent two wind speed values in location i, and $d = |w_s - w_t|$ their distance, then:

$$\begin{aligned}
\mathrm{Cor}(d) &= \mathrm{Cor}(f(w_s, i), f(w_t, i)) \\
&= \frac{1}{2^{\nu-1}\Gamma(\nu)}(\kappa d)^{\nu} K_{\nu}(\kappa d), \quad \alpha = \nu + d/2,
\end{aligned}$$

Details on smooth components

- Additionally a uniform correlation matrix is assumed across locations $j, k \in \{1, \ldots, I\}$. That is:

$$\mathrm{Cor}(f(w, j), f(w, k)) = \rho, \quad j \neq k$$

  which helps to pool information across locations, benefiting locations with less data.

Power estimate

This leads to the following power estimate at time t and location i:

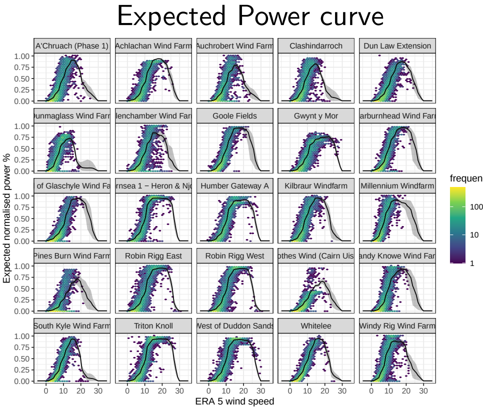
$$\hat{P}_{it} = \begin{cases} 0, & \text{with probability } \pi_{it}, \\ \mu_{it}^{+}, & \text{with probability } 1 - \pi_{it}. \end{cases}$$

Which leads to the expected power estimate:

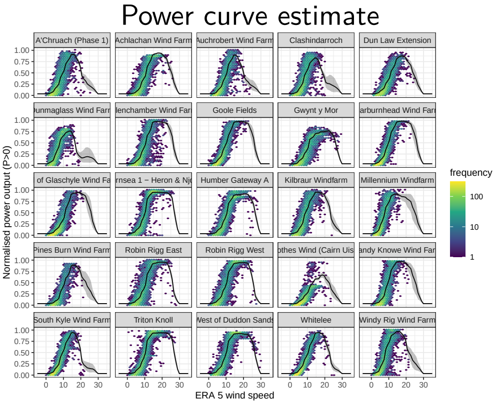
$$E[P_{it} | w_{it}, i] = (1 - \pi_{it}) \times \mu_{it}^{+}$$

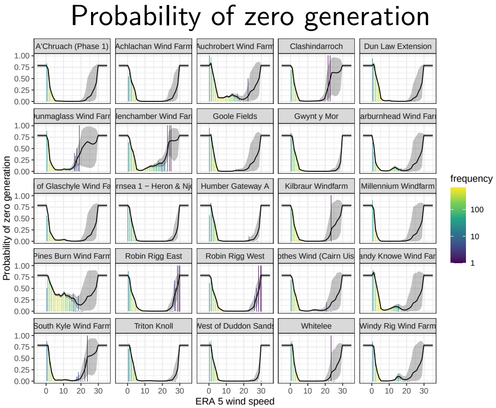
Power curve model estimates

- Starting models with
  - 1Y of data (2024)
  - 25 wind farms
  - Hourly data
  - Excluding outages listed in REMIT



Power curve model estimates



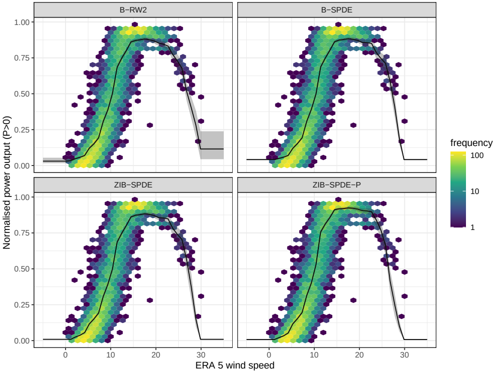


Alternative versions

- Beta model with RW2 power curve (B-RW2)
- Beta model with 1D SPDE power curve (B-SPDE)
- ZIB model with 1D SPDE power curve (ZIB-SPDE)
- ZIB model with 1D SPDE power curve penalised (ZIB-SPDE-P)

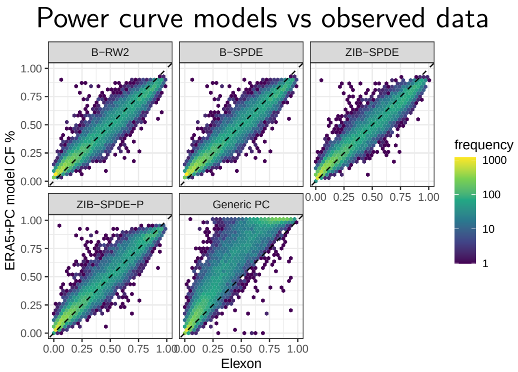
Zero Inflated models better capture low wind power regions.

Power curve estimates under different models



ZIB model vs observed data

- Power curve modelling highly improves the calibration
- ZIB models deal better with the tail regions
- Penalised version actually pulls estimate toward Generic PC
- We could pull estimate instead to a 5 parameter curve estimate



Error metrics

Error metrics calculated on 30% of the sampled wind farms data.

model	RMSE	MAE	Bias
Generic PC	0.2188	0.1583	0.1469
B-RW2	0.0995	0.0723	0.0191
B-SPDE	0.0994	0.0722	0.0190
ZIB-SPDE	0.0976	0.0691	0.0103
ZIB-SPDE-P	0.1016	0.0694	0.0269

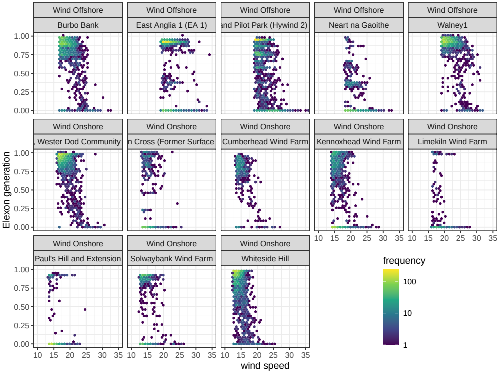
Pooling data for cut-off estimation

- Not enough data to estimate at wind farm level
- Even using all 5 years of data (2019-2024), only a few farms show cut-off behaviour

Pooling data at high wind speeds

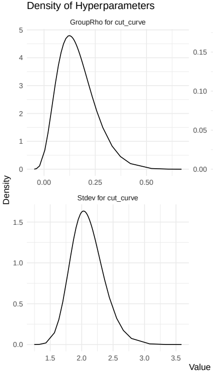
wind speed	prop. P=0	prop. P<=0.02	n obs.
[0,15]	0.22	0.23	3292
(15,20]	0.13	0.14	7062
(20,25]	0.11	0.13	5266
(25,30]	0.43	0.48	670
(30,Inf]	0.76	0.79	63

Tail behaviour of power curve

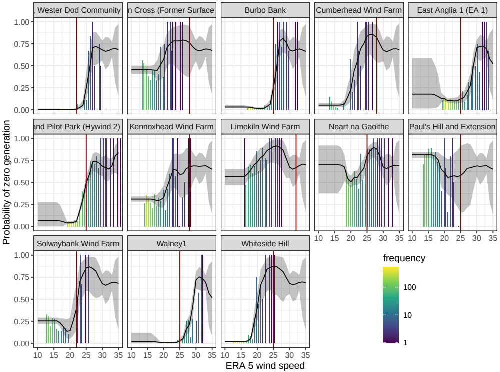


Model to estimate cut-off by wind speed

Hyperparameters of the cut-off model



Probability of zero generation at high wind speeds



Power Curve fitting

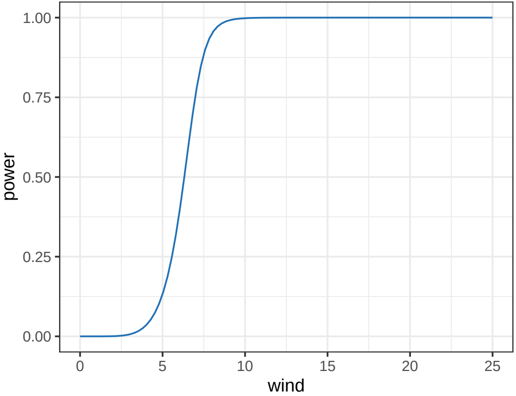
Classic parametric shape

The five-parameter formula (Lydia et.al, 2013):

$$P(u) = D + \frac{A - D}{(1 + (u/C)^B)^G}$$

where: A and D are the upper and lower asymptotes, C controls the inflection point, B is related to the slope at inflection point, and G controls the asymmetry.

5 parameters power curve

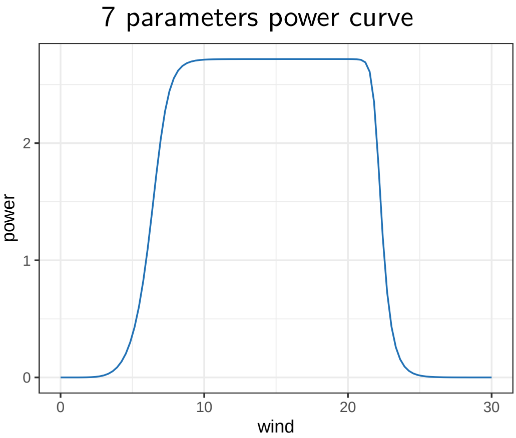


Parametric shape for cut-off

Adding extra parameters we can model de decay after cut-off:

$$P(u) = D + \frac{A - D}{(1 + (u/C_1)^{B_1})^G (1 + (u/C_2)^{B_2})^G}$$

where: A and D are the upper and lower asymptotes, C_1 is the first inflection point, C_2 is related to the cut-off, B_1 is related to the slope at inflection point, B_2 to the slope during cut-off, and G controls the asymmetry.

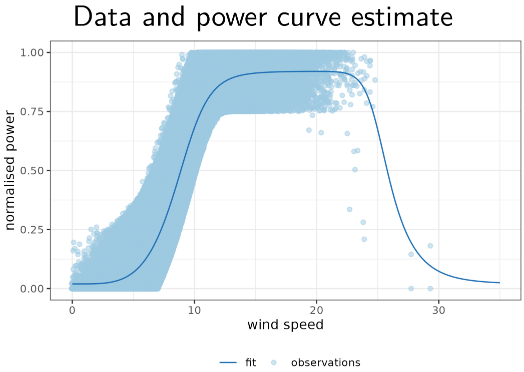


Parametric fit for Scottish wind farms

- We can fit the parametric shape using optim
- As long as there is a reasonable amount of points showing the cut-off optim can converge

Estimates from data

par	estimate
A	0.92
B1	-9.74
C1	10.06
D	0.02
G	0.43
B2	35.64
C2	24.92



Next Steps

- Spatiotemporal calibration model
- Non-stationary covariance model
- Comparison against benchmarks

Data sources

Wind speed

- ERA5 at wind farms
  - Hourly data
  - Spatial resolution 0.25° x 0.25°
  - 10m and 100m heights

Wind power

- Elexon BMU data (since 2019)
  - Half hourly data
  - Generation, curtailment, potential, capacity
  - Outage data (REMIT)
- REPD database
  - Location, turbine height, capacity

Previous research

Power curve modelling

Binning method

$$P_i = \frac{1}{n_i} \sum_{j=1}^{n_i} P_{ij}$$

where: P_{ij} is the j th power observation in bin i and n_i no. of observations in bin i

Logistic

$$P(u) = a \frac{1 + m \exp(-u/\tau)}{1 + n \exp(-u/\tau)}$$

where a represents the upper asymptote, n, m shape the lower asymptote, and τ controls the transition.

Power curve modelling

5 parameter curve

$$P(u) = D + \frac{A - D}{(1 + (u/C)^B)^G}$$

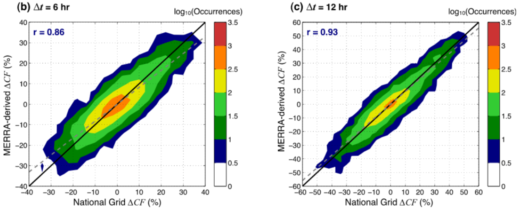
where: A and D are the upper and lower asymptotes, C is the inflection point, B the slope at inflection point, adn G controls the asymmetry.

Reanalysis data to quantify extreme wind power statistics

D. Cannon, D. Brayshaw, et. al (2015)

- MERRA wind speed validated with MIDAS
- Vertical interpolation with a logarithmic change
- Calibrated power curves based on manufacturers PC
- Use that to analys extreme low and high levels, and ramps

ΔCF comparison

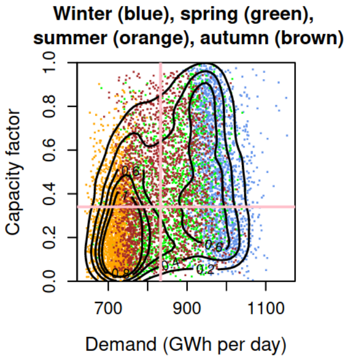


Balancing energy

H. Thornton, D. Brayshaw (2020) analyse the relationship between weather, energy demand, and wind power.

- ERA Interim wind speed
- Cubic power curve with air density correction
- CF compared with GB average from other studies
- Seasonal effects on Demand and

ΔCF comparison

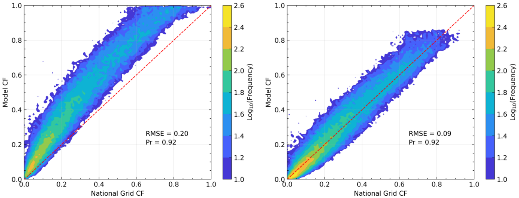


Analysis of extreme wind droughts

Panit Potisomporn, C. Vogel (2024) perform a extreme value analysis of wind droughts in GB.

- Use ERA5 wind speeds calibrated to MIDAS with QM
- Build a ML algorithm that learns how to extrapolate wind speed from 10m to hub height
- Use a 5 parameter logistic function to model power curve
- Model energy losses with factors by type.

CF calibration



Challenges in spatial wind calibration

Challenges in spatial wind calibration

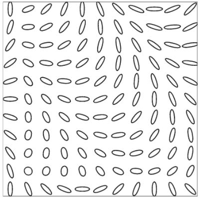
- Non-stationarity
  - (space) spatially varying coefficients
  - (wind direction) anisotropy
  - (time) regimes / storms and other weather events
- Physically informed model
  - Regularisation / penalisation
- Data dimensionality reduction
  - Aggregate time dimension (day)

Non-stationarity approaches found in the literature

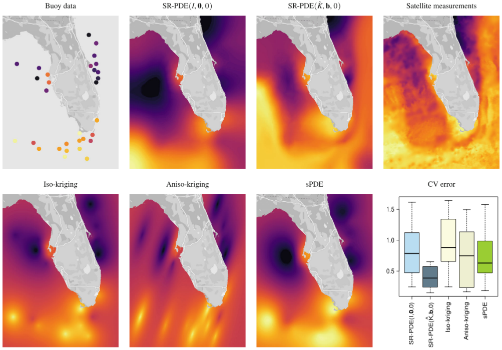
Regression with PDE regularisation

- Tomasetto et al.(2024) model anisotropy through a physics informed regression model
- Includding physical knowledge through a PDE regularisation term

Anisotropic non-stationary diffusion



Sea Surface Temperature (SST)



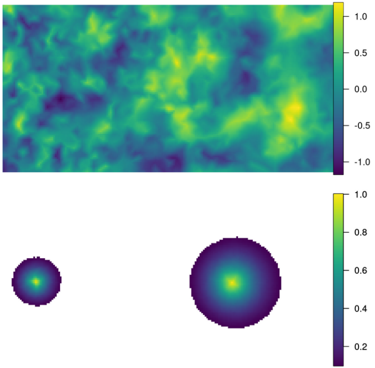
Explanatory variables in the covariance

- Ingebrigtsen, Lindgren, and Steinsland (2014)
- Regression on ranke and precision parameters of spatial term

$$\log \tau(\mathbf{s}) = b_0^{(\tau)}(\mathbf{s}) + \sum_{k=1}^{p} b_k^{(\tau)}(\mathbf{s})\,\theta_k, \qquad (2)$$

$$\log \kappa(\mathbf{s}) = b_0^{(\kappa)}(\mathbf{s}) + \sum_{k=1}^{p} b_k^{(\kappa)}(\mathbf{s})\,\theta_k. \qquad (3)$$

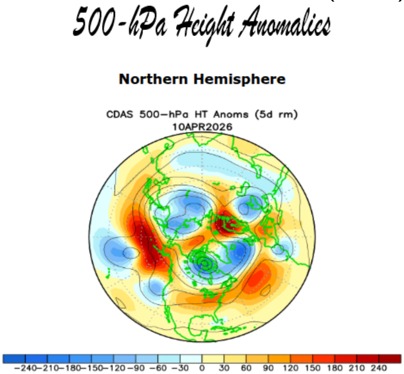
Field and correlation at two points



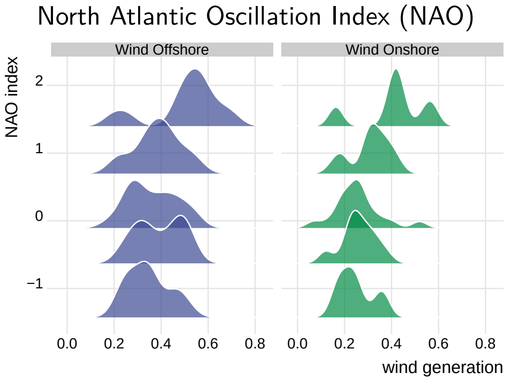
Variables relevant to non-stationarity in wind

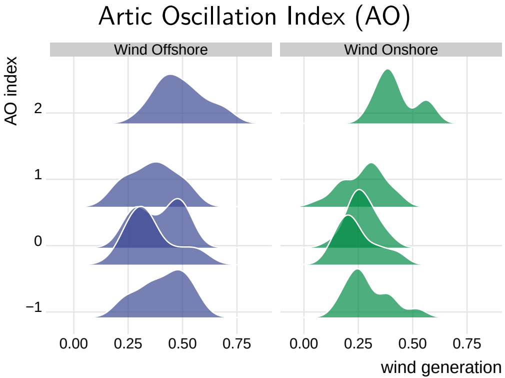
- Wind dynamics are non-stationary in space and time
- Weather regimes, storms, other events
- Circulation patterns (NAO, AO, EA, SCAN)
- NAO + leads to higher wind speeds UK and more variability

North Atlantic Oscillation (NAO)



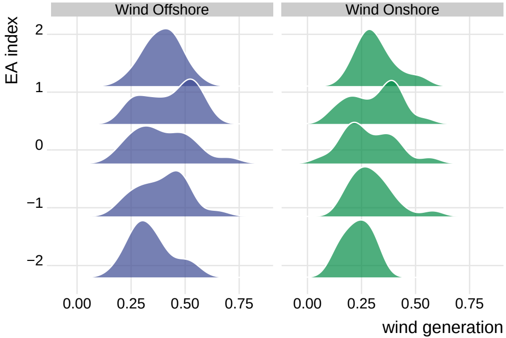
Cirtulation patterns and wind power



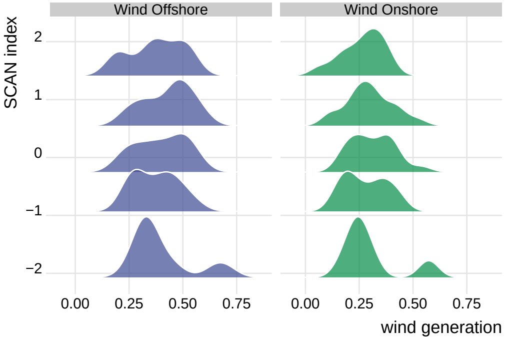


Cirtulation patterns and wind power

East Atlantic Index (EA)

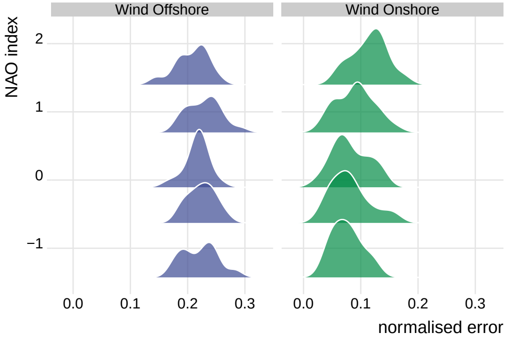


Scandinavian Oscillation Index (SCAN)

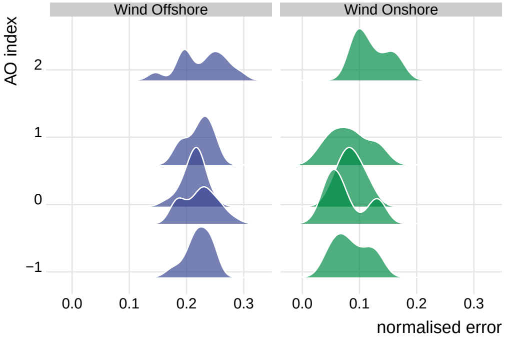


Cirtulation patterns and error in wind power estimates

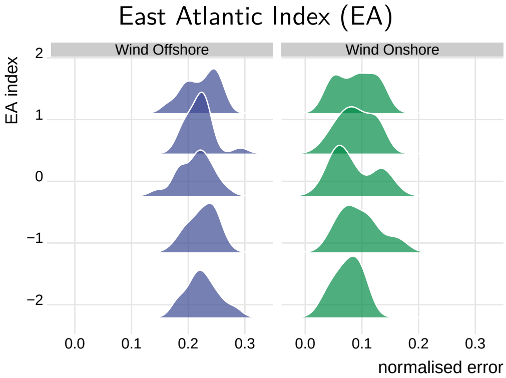
North Atlantic Oscillation Index (NAO)

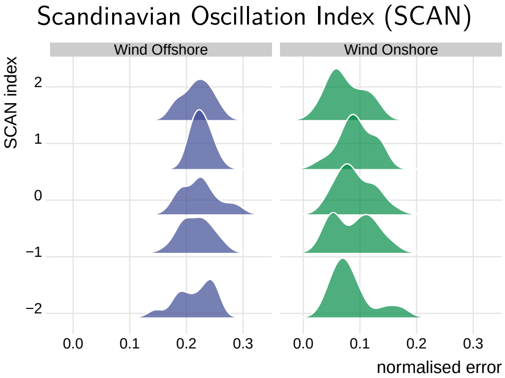


Artic Oscillation Index (AO)



Cirtulation patterns and error in wind power estimates





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